SQL QUERYING & DATA INSIGHTS REPORT

Program: AnalystLab Africa Data Analytics Internship | Week 3 | Database: Chinook Music Store

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This technical deliverable provides a thorough code documentation and analytical insight review of the SQL production scripts executed against the Chinook Relational Database model using Microsoft SQL Server (T-SQL). The analysis covers structural definitions, core aggregations, complex logical joins, window functions, and physical table optimization architectures.

1. Base Database Exploration & Schema Discovery

Query Operations:

- `SELECT TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_TYPE = 'BASE TABLE';`
- `SELECT TOP 5 * FROM Customer;` (Executed iteratively across all 11 system entities, including `Invoice`, `InvoiceLine`, `Track`, `Album`, `Artist`, `Genre`, `Employee`).

Why it Matters: It acts as the foundational structural audit. Mapping metadata through the system schemas allows analysts to explicitly determine table shapes, pinpoint data type distributions, and discover logical key relationships (e.g., matching primary keys to foreign keys) before compiling intensive multi-table join scripts.

Data Observations & Insights: The environment maps a clean, normalized relational digital storefront architecture spanning exactly 11 base tables. Crucial constraints are identified early: `CustomerId` stands as the primary structural bridge linking the `Customer` catalog details directly to transactional financial tables like `Invoice`, establishing a clear 1-to-many business behavior pattern.

2. Core Metrics, High-Value Filters, & Segment Aggregations

A. High-Basket Isolation

Query Operations: Filtered transaction records tracking only rows where `Total > 10`, ordered descending by the financial value field.

Why it Matters: It serves as a margin-optimization filter, allowing corporate strategy teams to immediately strip out minor single-track micro-purchases ($0.99 orders) and focus exclusively on high-basket transactions that move net margins.

Data Observations & Insights: Individual high-tier order checkouts strictly hit a ceiling in the $20 to $26 variance range, led prominently by Invoice 404 (Czech Republic) evaluating at `$25.86` and Invoice 299 (USA) evaluating at `$23.86`. This clearly demonstrates that core high-tier users purchase in bundles or full-length product offerings rather than disconnected single items.

B. Geographic Performance & Aggregated Group Filtering

Query Operations: Grouped global sales using `GROUP BY BillingCountry` to calculate `SUM(Total)`, `AVG(Total)`, and `COUNT(InvoiceId)`, subsequently evaluated using a post-aggregation `HAVING SUM(Total) > 100` clause.

Why it Matters: It centralizes regional intelligence, identifying core target geographies and providing an automated framework via the `HAVING` keyword to drop low-performing or minor marketing territories from high-level management views.

Data Observations & Insights: The **United States represents the absolute corporate revenue anchor**, generating a high-velocity `$523.06` across `91 total orders`, followed closely by Canada at `$303.96`. When the restrictive $100 baseline check is run, smaller international segments like Portugal ($77.24), India ($75.26), and Chile ($46.62) are automatically dropped from the active dashboard layer, proving corporate revenues rely extensively on a condensed North American and Western European customer cluster.

3. Advanced Joins, Personas, & Subquery Benchmarking

A. Inner Joins for Customer Personalization

Query Operations: Bound the `Customer` entity to the `Invoice` entity utilizing an explicit `INNER JOIN` linked via matching `CustomerId` elements, concatenating text string expressions to produce a clean `CustomerName` layout.

Why it Matters: It transforms abstract numerical identifier records into distinct human customer records, connecting financial accounting figures directly to real customer personas for client relationship management (CRM).

Data Observations & Insights: Top high-tier shoppers display high global diversity. The absolute highest transactional cart belongs to customer **Helena Holý** (Czech Republic) at `$25.86`, immediately followed by **Richard Cunningham** (USA) at `$23.86`.

B. Dynamic Baseline Evaluation via Subqueries

Query Operations: `SELECT * FROM Invoice WHERE Total > (SELECT AVG(Total) FROM Invoice);`

Why it Matters: It eliminates human bias or arbitrary hardcoded target values. By nesting a dynamic statistical calculation directly inside the `WHERE` statement, the query automatically shifts its baseline benchmark over time as the database grows.

Data Observations & Insights: This query perfectly segments regular transactional traffic away from premium, above-average basket sizes, giving operations a dynamic, real-time list of high-value client acquisitions.

4. Analytical Window Functions & Market Tier Rankings

Query Operations: Utilized a Common Table Expression (CTE) tracking lifetime customer spend values, paired with the analytical window function: `RANK() OVER (PARTITION BY Country ORDER BY TotalSpent DESC)`.

> **Why it Matters:** Standard grouping filters always allow high-volume powerhouse countries (like the USA) to drown out crucial client trends in smaller, developing regional hubs. Partitioning by geographic boundaries isolates the localized VIP spenders within each individual nation.
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> **Data Observations & Insights:** In the highly competitive Latin American market territory, **Diego Gutiérrez** takes the undisputed #1 spot for Argentina with a lifetime valuation of `$37.62`. Concurrently, in Brazil, **Luís Gonçalves** takes the #1 rank at `$39.62`, while Roberto Almeida, Alexandre Rocha, and Eduardo Martins are locked in a structural three-way tie for the #2 spot at exactly `$37.62` each. This provides localized regional teams with precise customer lists for target-driven loyalty reward deployments.

5. Operational Optimization & Inventory Velocity Metrics

A. Catalog Engine Tracking

Query Operations: Linked cross-relational mapping tables (`InvoiceLine` ➔ `Track` ➔ `Genre`) to isolate unit volumes and sum gross track-level financials via a restricted `TOP 10` filter.

> **Why it Matters:** It tells corporate management exactly which product assets function as the true velocity engines of the company, showing where to direct specific promotion or marketing budgets.
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> **Data Observations & Insights:** The track asset titled **"Dazed And Confused"** categorized under the **Rock** genre represents a key volume product, driving `$4.95` in total revenue across `5 unique customer units sold`. The wider Rock category shows consistent repeat item purchase behavior across the entire transaction database.

B. Physical Table Optimization via Database Indexes

Query Operations: Executed a physical database structural change: `CREATE INDEX idx_customer_country ON Customer(Country);` and verified execution via a catalog query to the `sys.indexes` system table view.

> **Why it Matters:** In a production environment scaling across millions of rows, running heavy filters or regional Joins causes massive performance lag. Building a physical B-Tree index prevents slow, computationally expensive "table scans," optimizing analytical query response times down from minutes to milliseconds.